
SENSATION & PERCEPTION — COMPLETE OPEN-BOOK MASTER NOTE



1. INTRODUCTION: How We Experience the World

Your senses work **as a team** to help you navigate the environment.

Two main processes:

1. **Sensation** — receiving raw information.
2. **Perception** — interpreting the information to make sense of the world.



2. SENSATION

Definition:

Sensation is the **process by which sensory receptors and the nervous system detect and receive stimulus energy** from the environment.

How Sensation Works (Step-by-step):

1. A **physical stimulus** (light, sound, smell, touch, taste) hits a sensory organ.
2. Sensory receptors **absorb** this energy.
3. They **convert** the energy into **neural impulses** (electrical signals).
4. These impulses travel to the brain through neural pathways.

Key Idea:

Sensation = **Raw detection of stimulus**

It does **not** yet have meaning.



3. PERCEPTION

Definition:

Perception is the **process of selecting, organizing, and interpreting sensory information** to form meaningful understanding of objects and events.

How Perception Works:

- The brain receives signals from sensation.
- It **organizes** (groups, separates, prioritizes).
- It **interprets** them based on experience, memory, and expectations.

Key Idea:

Perception = **Meaning-making**

Without perception, sensations are just meaningless signals.



4. HOW SENSATION & PERCEPTION WORK TOGETHER

Flow:

Stimulus → Sensory Organ → Sensation → Neural Messages → Brain → Perception → Understanding

Example:

- Sensation: Light hits the retina.
- Perception: Brain recognizes a face.

Diagram (text version):

Stimulus → Receptors → Neural Signals → Brain → Interpretation → Perception



5. THRESHOLDS

Threshold = The dividing line between detectable and undetectable stimulation.

5.1 ABSOLUTE THRESHOLD

Definition:

The **minimum** amount of stimulation needed to detect a stimulus **50% of the time**.

Examples:

- Seeing a candle flame from 30 miles away on a dark night.
 - Hearing a watch tick from 20 feet away.
 - Smelling one drop of perfume in a three-room apartment.
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5.2 DIFFERENCE THRESHOLD (Just Noticeable Difference – JND)

Definition:

The **smallest** change in stimulus intensity that a person can detect.

Important Rule — Weber's Law:

The stronger the original stimulus, the greater the change needed to notice a difference.

Examples:

- 5 → 10 lbs = noticeable
- 100 → 105 lbs = not noticeable
- TV volume 10 → 12 = noticeable
- TV volume 70 → 72 = barely noticeable

Difference threshold deals with **change**.



6. SIGNAL DETECTION THEORY (SDT)

(Very important for exam questions)

Purpose:

Explains how and why people detect faint signals differently depending on psychological and physical conditions.

Key Concepts:

- The brain always has **neural noise** (background random firing).
- A faint stimulus creates a **signal**.
- The brain must decide:

“Is this noise or is this a real signal?”

Detection depends on:

1. **Stimulus intensity**
2. **Person’s state:**
 - fatigue
 - stress
 - expectation
 - motivation
 - alertness

Outcomes (SDT matrix):

Actual Signal Response		Example
Present	Hit	hearing a sound that is actually there
Present	Miss	failing to hear someone call your name
Absent	False Alarm	thinking your phone vibrated when it didn’t
Absent	Correct Rejection	recognizing that a sound was not real

Application Areas:

- Decision-making
- Medical diagnosis
- Security checking
- Radar operation
- Military alert systems



7. SENSORY ADAPTATION

Definition:

Reduced sensitivity to a constant or unchanging stimulus.

Why it happens:

The brain filters out irrelevant, constant information to focus on **new and important changes**.

Examples:

- Becoming unaware of your clothes touching your skin.
 - No longer smelling perfume after a few minutes.
 - Chef not noticing kitchen heat.
 - Worker ignoring construction noise.
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**8. VISION: HOW THE BRAIN MAKES SENSE OF SIGHT****Key Idea:**

Vision is **not** just receiving light — it is the **brain's interpretation** of that light.

Why illusions occur:

Our brain evolved shortcuts to interpret the world quickly. These shortcuts work 99% of the time, but sometimes fail → illusion.

**9. THE VISUAL PROCESS (STEP-BY-STEP)****Text Diagram:**

Light → Cornea → Pupil → Lens → Retina → Optic Nerve
→ Thalamus → Visual Cortex (Occipital Lobe)

Important parts:

- **Retina:** Converts light into neural signals.
- **Optic Nerve:** Sends signals to brain.

- **Visual Cortex:** Processes shapes, colors, movement, faces.
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10. COLOUR VISION

Young–Helmholtz Trichromatic Theory

Main Idea:

The retina has **three types of cones**:

- **Red**
- **Green**
- **Blue**

All perceived colors = combinations of these three.

Examples:

- Red + Green = Yellow
 - Green + Blue = Cyan
 - Red + Blue = Purple
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11. COLOR BLINDNESS

Definition:

Reduced ability to see or distinguish between certain colors.

Most common type:

Red–Green color blindness.

Slide Example:

People with normal vision see “74” in the dot pattern.

People with color blindness may not.



12. VISUAL INFORMATION PROCESSING

12.1 Feature Detection (Hubel & Wiesel)

Definition:

Specialized neurons in the occipital lobe respond only to specific visual features:

- Edges
- Angles
- Lines
- Movements

These neurons help identify objects quickly.

12.2 Supercell Clusters (David Perrett)

Some brain cells respond to:

- Gaze direction
- Head angle
- Posture
- Body movement

These cells help:

- read social cues
 - understand others' attention
 - detect threats
 - predict movement
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13. PARALLEL PROCESSING

Definition:

The brain processes **multiple visual aspects at the same time**.

The four parts processed in parallel:

1. **Color**
2. **Motion**
3. **Form**
4. **Depth**

Example:

When watching a bird, you instantly know:

- its color
- shape
- how fast it's moving
- how far away it is

Your brain does all these **simultaneously**.

**14. SIMPLIFIED SUMMARY OF VISUAL PROCESSING****Memory Flowchart:**

Retina detects light →

Feature detectors analyze edges and motion →

Supercells integrate movement + direction →

Parallel processing handles color, form, depth, motion →

Brain combines everything into meaningful vision

**EXAM SURVIVAL SUMMARY (90-SECOND REVIEW)**

- **Sensation** = detecting physical energy

- **Perception** = interpreting it
 - **Absolute Threshold** = minimum detectable 50%
 - **Difference Threshold (JND)** = smallest noticeable change
 - **SDT** = detection depends on stimulus + person's state
 - **Sensory Adaptation** = lower sensitivity after constant stimulus
 - **Vision Pathway** = Retina → Brain
 - **Trichromatic Theory** = 3 cones (RGB)
 - **Color Blindness** = difficulty distinguishing colors
 - **Feature Detection** = specialized cells for edges, lines
 - **Supercells** = gaze, posture, movement
 - **Parallel Processing** = color, motion, form, depth simultaneously
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